

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Total Alkalinity	WQP 015	1.5	

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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC
1.1	8/30/23	Included correction for μM to $\mu\text{mol/kg}$ in SOP, change sample bottle to 150 mL Pyrex bottles, and changed from analyzing at 20°C to room temperature	TAC
1.2	10/2/23	Changed SOP so that all samples are analyzed in μM and converted to $\mu\text{mol/kg}$ in post analysis and updated bottle cleaning procedure for screw-top bottles.	TAC
1.3	10/4/23	Corrected $\mu\text{mol/kg}$ to μM conversion window text.	TAC
1.4	11/22/23	Updated method for pH electrode with thermometer and revised CRM accuracy section for clarity.	TAC
1.5	3/19/24	Minor edits to improve method clarity, increased recommended sample volume to 25 mL to improve instrument precision, and reduced precision threshold to a maximum of $\pm 4 \mu\text{M}$ based on instrument performance.	TAC

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1. Title

Standard Operating Procedure for Total Alkalinity

2. Applicability

This SOP applies to the determination of seawater Total Alkalinity (TA) at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1 Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater surface and bottom seawater TA at shallow coral reef sites around Puerto Rico.
- 3.2 Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater TA data following the Quality Project Plan.

4. Method

Standard Method from Dickson et al. (2007).

5. Health and Safety

- 5.1 Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2 Read each reagent SDS prior to use for safe handling and disposal.
- 5.3 Wear disposable latex gloves and safety glasses while performing this procedure.

6. General Precautions

- 6.1 Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2 Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the line does not become entangled with the boat propeller.
- 6.3 Do not use solvents, alkaline or abrasive cleaners or other abrasives such as steel wool or scouring powder.
- 6.4 Keep temperature constant throughout the system to avoid bubbles.

7. Training

- 7.1 Personnel shall be trained in the proper sample collection, handling and analysis procedure for TA.
- 7.2 Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

8.1 Sampling Equipment

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- 8.1.1 Van Dorn Bottle
- 8.1.2 Sampling line
- 8.1.3 Van Dorn messenger
- 8.1.4 Tygon Tubing

8.2 Consumables

- 8.2.1 Plastic drawing tubes
- 8.2.2 150 mL Pyrex Screw-top Bottles
- 8.2.3 Glass Storage Bottle (25 mL)
- 8.2.4 Glass Storage Bottle HCl (500mL)
- 8.2.5 Glass beaker for pH calibration (25 mL)
- 8.2.6 Pipette (10-100µL)
- 8.2.7 Pipette Tips (100µL)
- 8.2.8 Stand for the electrode
- 8.2.9 pH 4, 7, and 10 calibration solutions (stored at 20°C in BOD incubator)
- 8.2.10 Distilled water
- 8.2.11 Mercuric (II) chloride

8.3 Total Alkalinity Gran Titration System

- 8.3.1 AS-ALK3 Alkalinity Unit
- 8.3.2 TALK software
- 8.3.3 Orion Star A211 Benchtop pH Meter
- 8.3.4 ROSS Orion pH electrode
- 8.3.5 Titration vessel
- 8.3.6 Insulated beakers
- 8.3.7 Stir plate
- 8.3.8 Magnetic stirrer
- 8.3.9 Certified Reference Material for Oceanic CO₂ measurements

8.4 Chemicals & Solutions Preparation

Compound	CAS No.		Safety classification
Mercuric chloride	7487-94-7	GHS03, GHS05, GHS06, GHS07, GHS09	Danger - Oxidizing, Warning/Danger-Corrosive, Corrosive, Irritant, Danger – Toxic, Warning/Danger- Environmentally Damaging.
Hydrochloric acid, 0.1 N, HCl	7647-01-1	GHS05, GHS07	Warning/Danger – Corrosive, Irritant
Sodium Chloride	7647-14-5	–	Not a hazardous substance
Sodium carbonate	497-19-8	GHS07	Irritant

- 8.4.1 HCl (0.09661 N)+ NaCl (0.6 N) solution

Reagent	Amount	Preparation Instructions
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Hydrochloric acid HCl (0.1N)	500 mL	1. Weigh 17.53 NaCl g of sodium chloride (0.6N). 2. Measure 500 mL 0.1 N HCl in a graduated cylinder. 3. Add the 17.53 g of NaCl and 500 mL 0.1 N HCl to a 500 mL Pyrex bottle. 4. Mix thoroughly and label “HCl (0.09661 N)+ NaCl (0.6 N) solution”
Sodium Chloride NaCl (0.6N)	17.53 g	
General Details		
Storage	500 mL glass bottle at room temperature	
Stability	Indefinitely, verify with Certified Reference Material	
Reagent Label	N/A	

8.4.2 Mercuric (II) chloride solution

Reagent	Amount	Preparation Instructions
Mercury (II) Chloride (HgCl ₂)	2.0 g	1. Add the 2 g HgCl₂ and 25 mL H₂O to a 25 mL Pyrex bottle. 2. Close bottle, shake to mix thoroughly, and label “Danger HgCl₂ solution”. 3. Place in a sealed plastic bag.
Distilled water	25.0 mL	
General Details		
Storage	25 mL glass bottle at room temperature in a sealed plastic bag	
Stability	Indefinitely	
Reagent Label	Warning, Danger, Toxic	

8.4.3 Sodium carbonate solution (2300 µM)

Reagent	Amount	Preparation Instructions
Sodium carbonate (Na ₂ CO ₃)	0.061 g	1. Weigh 0.061 g of sodium carbonate. 2. Measure 500 mL distilled water.

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Distilled water	500 mL	3. Add sodium carbonate and distilled water to a 500 mL glass bottle and shake to mix. 4. Label “2300 $\mu\text{M Na}_2\text{CO}_3$ ”
General Details		
Storage	500 mL glass bottle at room temperature	
Stability	Indefinitely, verify with Certified Reference Material	
Reagent Label	N/A	

9. Procedure

9.1 Sample Collection

- 9.1.1 Prepare a Van Dorn bottle to be deployed.
- 9.1.2 Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3 Drop the bottle messenger to close the Van Dorn bottle at the desired depth.
- 9.1.4 Pull the Van Dorn bottle back up into the boat.
- 9.1.5 Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6 Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7 Place the Tygon tubing at the bottom of the inverted 150 mL screw-top Pyrex glass bottle to rinse with water from the Van Dorn bottle.
- 9.1.8 Flip the sample bottle opening side up to begin filling the sample bottle from the bottom.
- 9.1.9 Collect the sample until the bottle overflows 1.5 times the sample volume.
- 9.1.10 Pinch the Tygon tubing to stop flow and remove the tubing from the sample vial. This will leave approximately 1% head space in the bottle.
 - 9.1.10.1 Refer to Appendix J for the seawater sample collection order.
- 9.1.11 Poison the sample.
 - 9.1.11.1 Add 50 μL mercury (II) chloride (HgCl_2) to the 150 mL sample bottles.
 - 9.1.11.2 Larger sample bottles can be used, but the sample should be poisoned at the same proportion of mercury (II) chloride to sample.
- 9.1.12 Ensure that the sample is filled to the top of the bottle and screw the lid on tightly to prevent any gas exchange.
- 9.1.13 Gently invert the bottle several times to mix the mercuric chloride with the sample.
- 9.1.14 Place the sample bottle inside the cooler until analysis in the lab.

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9.2 Instrument Start-up

- 9.2.1 Turn the equipment ON in the following order:
 - 9.2.1.1 Laptop
 - 9.2.1.2 Apollo AS ALK3 titrator
 - 9.2.1.3 pH meter
- 9.2.2 Open the TALK v1.0T software (icon at laptop's Desktop).
- 9.2.3 Verify communication ports near the left upper corner of the main screen.
 - 9.2.3.1 Pump = COM 1
 - 9.2.3.2 pH = COM 2
- 9.2.4 Click on the "Initialize" button under the Port COM selection.
- 9.2.5 Input the datafile name "Project_YYYYMMDD".
- 9.2.6 Save it in the "Alkalinity" folder under the Documents file.
- 9.2.7 Verify the status fields on the upper right hand corner of the window display "Ready".
- 9.2.8 Connect the instrument's acid tubings into the calibration acid and waste container to flush the titration acid.
 - 9.2.8.1 Place the black rubber stopper into the titration acid container lid to reduce evaporation.
 - 9.2.8.2 Place the acid inlet line into one of the titration acid holes and ensure that it is fully submerged in the acid.
 - 9.2.8.3 Cover the second open hole on the rubber stopper with parafilm.
 - 9.2.8.4 The acid outlet line should be placed in the syringe holder at the top of the acid waste container.
 - 9.2.8.5 Press "OK" when ready to start flushing HCl.
 - 9.2.8.6 If bubbles are observed in tubing, flick the Tygon intake and outlet tubing to ensure all bubbles are removed from the acid tubing.
- 9.2.9 Click "Flush Acid Syringe" x 1 to flush the acid syringe again.
 - 9.2.9.1 If bubbles are observed in tubing, flick the Tygon intake and outlet tubing to ensure all bubbles are removed from the acid tubing.
 - 9.2.9.2 Repeat step 9.2.9 as necessary until there are no bubbles in the acid Tygon tubing.
- 9.2.10 Connect the instrument's sample tubings into the sample and waste container to flush the sample lines.
 - 9.2.10.1 Place the sample inlet line into one of the 2300 μ M sodium carbonate solution and ensure that it is fully submerged in the solution.
 - 9.2.10.2 The sample outlet line should be placed in the syringe holder at the top of the sample waste container.
- 9.2.11 Click "Flush Sample Syringe" x 1 to flush the sample syringe.
 - 9.2.11.1 If bubbles are observed in tubing, flick the Tygon intake and outlet tubing to ensure all bubbles are removed from the sample tubing.

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- 9.2.11.2 Repeat step 9.2.10 as necessary until there are no bubbles in the sample Tygon tubing.
- 9.2.12 Prepare the pH electrode for analyses.
 - 9.2.12.1 Remove the blue plug covering the electrode refill opening.
 - 9.2.12.2 Check the electrode fill level.
 - 9.2.12.3 Fill the electrode up to the level of the opening using Ross Reference Electrode Filling Solution as needed.
 - 9.2.12.4 Remove the pH electrode from the storage solution.
 - 9.2.12.5 Rinse the pH electrode with distilled water and blot dry with a KimWipe.
 - 9.2.12.6 Be careful not to damage the glass bulb of the pH electrode.
- 9.2.13 Ensure that the Temperature (°C) in the Sample Information Input section is set to the temperature measured by the pH meter in the sample. Click Save if any values were changed.
- 9.2.14 Check that the Sample Volume in the Sample Information Input Section is set to 25 ml. Click Save if any values were changed.
 - 9.2.14.1 Note: Smaller sample volumes may be used if needed but typically result in lower instrument precision.
- 9.2.15 Write your initials in the User section.
- 9.2.16 Write the Water Quality Project in the Project section.
- 9.2.17 Use a piece of Parafilm to attach the titration acid outlet line to the pH electrode so that the end of the tubing is about level with the end of the pH electrode.
- 9.2.18 The pH calibration (Section 10) and calibration acid standardization (Section 11) should be completed for each new batch of titration acid or whenever the titration acid fails to produce accurate results of certified reference materials.
 - 9.2.18.1 Otherwise, proceed directly to Section 9.3 Sample Analysis.

9.3 Sample Analysis

- 9.3.1 Place all samples to be analyzed at approximately room temperature for best results.
- 9.3.2 Input the sample volume in the “Sample Information Input” box.
- 9.3.3 Click “Save” to save current values.
 - 9.3.3.1 Click “Reload” to revert to last saved values.
- 9.3.4 Uncheck the box of S > 30 on the right side of the window and type in the salinity of the seawater sample.
- 9.3.5 Input “Salinity (PSU)” for known salinity samples.
- 9.3.6 Input “Sample Name” in the box below the salinity input.
- 9.3.7 Click the green button “Start New Analysis” to start titration for a new sample.
- 9.3.8 Confirm or make changes to the sample name, sample volume, and salinity in the pop-up window.

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- 9.3.9 Click “Start” when ready to start titration.
 - 9.3.9.1 When prompted by the software, place the sample tube into a clean titration vessel with a stir bar and click okay.
 - 9.3.9.2 Place the titration vessel with the sample onto the stir bar.
 - 9.3.9.3 Set the stirrer to the black line, approximately the third circle setting, this creates a vortex pattern without bubbles.
 - 9.3.9.4 Carefully lower the electrode with acid tubing into the titration vessel until reaching the stop point on the ring stand.
 - 9.3.9.5 Wait until the pH meter displays “Ready.”
 - 9.3.9.6 Click ok to start the titration when prompted by the software.
- 9.3.10 Wait for the program to display results.
- 9.3.11 Lift the pH electrode out of the solution and turn off the stirrer.
- 9.3.12 Rinse the pH electrode with distilled water and carefully blot dry to avoid damaging the glass bulb.
- 9.3.13 Empty the titration vessel into the sample waste container.
- 9.3.14 Rinse the vessel and stir bar with DI water and dry both with a KimWipe to prepare the vessel for the next analysis.
- 9.3.15 Ideally all samples should be analyzed at room temperature to minimize differences in density between samples.
 - 9.3.15.1 Record the temperature of each sample after TA analysis using the pH meter and record the Date, Time, Sample ID, Temperature, and TA (μM) in the TA laboratory notebook.
- 9.3.16 Prepare a new sample.
 - 9.3.16.1 Click either the green button “Start New Analysis” to begin a new sample or “Repeat Last Analysis” to repeat an analysis of the previous sample.
- 9.3.17 Samples should be analyzed in the following order:
 - 9.3.17.1 Run 3 sodium carbonate solution ($2300\ \mu\text{M}$) samples to ensure the system is precise ($\pm 3\ \mu\text{M}$ difference of each other).
 - 9.3.17.2 Troubleshoot instrument and run additional sodium carbonate samples until precision target is established before proceeding.
 - 9.3.17.3 Run 1 CRM to ensure the system is accurate ($\pm 4\ \mu\text{M}$ of stated value).
 - 9.3.17.3.1 The CRM TA values for the respective CRM Batch numbers can be accessed via:
https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/Dickson_CRM/batches.html
 - 9.3.17.3.2 **The CRM TA values are reported in $\mu\text{mol/kg}$ on the NOAA website and must be converted to μM before entering the value into the Apollo system software.**

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- 9.3.17.3.3 If the CRM is accurate to within $\pm 4 \mu\text{M}$, continue to step 9.3.17.6 to analyze samples.
- 9.3.17.3.4 If the CRM is not accurate to within $\pm 4 \mu\text{M}$, repeat CRM to determine if the system is precise.
- 9.3.17.3.5 If the repeated CRM are precise but not accurate, proceed to section 10 and 11 to recalibrate the pH electrode and HCl acid.
- 9.3.17.3.6 If the repeated CRM are not precise, troubleshoot the instrument using sodium carbonate solution ($2300 \mu\text{M}$) to determine the cause of precision.
- 9.3.17.4 Run 10 samples.
- 9.3.17.5 Run 1 x CRM to ensure the system is still accurate to within $\pm 4 \mu\text{M}$ after analyzing the samples.
- 9.3.17.6 Repeat the cycle of 10 samples followed by a CRM within $\pm 4 \mu\text{M}$ of the stated value and make sure that the day ends on a CRM that is within $\pm 4 \mu\text{M}$ of the stated value.
- 9.3.18 Perform the cleanup instructions described in Section 9.4 after running the last sample analysis.

9.4 Instrument Powering Off

- 9.4.1 Flush both syringes and pump valves with DI water.
 - 9.4.1.1 Click “Flush Acid Syringe”.
 - 9.4.1.2 Click “Flush Sample Syringe”.
 - 9.4.1.3 Click on the “x1” button to the left of each button to change it to “x2” to flush each syringe twice.
 - 9.4.1.4 In case air bubbles get trapped inside the syringe, you may repeat the flushing to get rid of them. If that does not work, refer to Section C7 of the user manual to remove the plunger from the syringe.
- 9.4.2 Click the “X” in the top right corner of the main window to close the program.
- 9.4.3 Confirm that both syringes have been cleaned with DI water on the pop-up window.
- 9.4.4 Click “OK” to close the program.
 - 9.4.4.1 Or click “Cancel” to return to the main window.
- 9.4.5 Close the electrode refill opening with the corresponding blue cover and place the electrode back into the storage solution container.
- 9.4.6 Power off the titrator.
- 9.4.7 Power off the pH meter.

10. Calibration of the pH electrode

- 10.1 Verify the calibration information in the software’s main screen.

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- 10.2 Select the “Recalibrate” button in the “Calibration Information” box.
- 10.3 Input the temperature of pH buffers in the “Sample Information Input” box.
- 10.4 Click “Save”.
- 10.5 Enter the pH standard values for 20°C for the calibration (pH = 4.00, 7.01, 10.02) in the appropriate space within the “Calibration Information” box.
- 10.6 Pour the calibration standard into a small glass beaker with a magnetic stirrer.
 - 10.6.1 Volume should be enough to cover the electrode junction.
- 10.7 Submerge the electrode in the first pH buffer.
- 10.8 Click on the “Read” button aligned with the corresponding pH buffer value to the right of the “Calibration Information” box.
 - 10.8.1 Notice that the EMF (mV) box displays the reading from the pH meter and the “Read” button becomes “Save”.
- 10.9 Notice the two numbers at the bottom right of the “Calibration Information” box start to count up indicating the time in seconds of a stable reading (top) and total time in seconds after clicking read (bottom).
- 10.10 Click “Save” once the upper number stabilizes.
 - 10.10.1 In case you need to repeat the measurements, you may click the circling arrows that will appear after saving the measurement.
 - 10.10.2 The pH buffer can be preserved in the polypropylene bottle for a subsequent calibration of the pH electrode for the TA system.
- 10.11 Repeat steps 10.6 to 10.11 for the remaining 2 buffers.
- 10.12 Click “Save” at the bottom of the box to save calibration.
 - 10.12.1 Repeat the calibration process if a pop-up window appears alerting the user of a low efficiency (<95%) electrode.
- 10.13 Follow instructions on Section 9.4 to standardize the HCl solution.
 - 10.13.1 Standardization should be performed in less than an hour of finishing the pH electrode calibration. The pH electrode will have to be recalibrated if more than an hour has elapsed.

11. Standardization of the HCl solution

- 11.1 Prepare HCl + NaCl titrant solution.
 - 11.1.1 Follow instructions in Section 8.4.1
- 11.2 Ensure accuracy and precision of the system.
 - 11.2.1 Follow steps 9.5.2 to 9.5.11 for instructions on how to run a sample.
 - 11.2.2 Run 3 sodium carbonate solutions to ensure the system is precise (± 3 $\mu\text{mol/kg}$ difference of each other).
 - 11.2.2.1 Troubleshoot instrument and run additional sodium carbonate samples until precision target is established before proceeding.
 - 11.2.2.2 Note: If precision has already been established for the day, proceed to step 11.3 to begin the HCl calibration using CRM.
- 11.3 Click on the green button “HCl Calibration” that appears near the bottom right of the main window.

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- 11.3.1 The program prompts to perform a pH electrode calibration first if the previous pH electrode calibration was performed more than an hour ago.
 - 11.3.1.1 Click “Yes” to proceed to calibration or “No” to start the HCl standardization.
- 11.4 Input calibration values in the pop-up window:
 - 11.4.1 Sample volume should be 25 ml although smaller samples may be used if needed.
 - 11.4.1.1 Note: Smaller sample sizes typically have lower instrument precision.
 - 11.4.2 ID of the Talk standard solution
 - 11.4.2.1 Certified Reference Material (CRM) for Total Alkalinity.
 - 11.4.2.2 CRM batch information is available at:
https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/Dickson_CRM/batches.html
 - 11.4.2.3 **The CRM TA values are reported in $\mu\text{mol/kg}$ on the NOAA website and must be converted to μM before entering the value into the Apollo system software.**
 - 11.4.3 Total alkalinity of the CRM (μM) at the input temperature.
 - 11.4.3.1 Press the circling arrows to access the TA unit conversion window.
 - 11.4.3.2 Input the reported TA of the CRM in $\mu\text{mol/kg}$.
 - 11.4.3.3 Input the Salinity of the CRM.
 - 11.4.3.4 Click “OK” to accept the calculated CRM in units of μM .
 - 11.4.4 Input the estimated concentration of the HCl solution.
 - 11.4.4.1 Note: if re-calibrating acid, the current acid concentration can be used. If calibrating newly prepared acid, input the estimated 0.09661 N HCl acid calibration as prepared in 8.4.1.
- 11.5 Click “OK” to save the parameters.
- 11.6 Verify the pump connections suggested in the pop-up window.
 - 11.6.1 Port B of the sample pump to the CRM.
 - 11.6.2 Port A of the sample pump to the Waste container.
- 11.7 Click “OK” and the system will rinse the pump and lines.
- 11.8 Move Port A of the sample solution to the titration vessel to collect sample for the titration when the pop-up window reminder appears.
- 11.9 Click “OK” when ready.
- 11.10 When prompted, turn on the stirrer.
- 11.11 Ensure the pH meter with acid output (Port A of the acid pump) is in the titration vessel.
- 11.12 Click “OK” when ready.
- 11.13 Monitor titration process on the multiple displays under the “Current Sample Results” section.
 - 11.13.1 The software will perform the corresponding calculations (end point acid volume and TALK) automatically and save it under C:\Talk\Titration Data.

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- 11.14 Click “Yes” to start a repeat of the titration.
- 11.15 Turn off the stirrer.
- 11.16 Remove the titration vessel from the water jacket beaker.
- 11.17 Rinse the electrode and the HCl outlet tubing with distilled water.
- 11.18 Gently blot dry the electrode with a Kimwipe.
- 11.19 Put a clean and dry vessel for the next titration.
- 11.20 Put the CRM solution outlet into the titration vessel and click “OK” to dispense the solution into the titration vessel.
- 11.21 Remove the standard solution outlet tubing from the vessel and follow the instructions on the window to place the vessel on the stirrer for analysis.
- 11.22 Turn the stirrer on to ensure safe operation of the stir bar prior to lowering the pH electrode into the titration vessel.
- 11.23 Lower the pH electrode and the HCl outlet tubing inside the vessel
 - 11.23.1 Caution not to hit the electrode with the bottom of the vessel or the stir bar.
- 11.24 Ensure everything is correct and click “OK” to start the titration.
- 11.25 After three titrations, the program calculates the average and relative standard deviation (RSD) of the calculated HCl concentrations.
- 11.26 If two of the CRM values are within 0.1% RSD, click “Accept” to save the HCl concentration.
 - 11.26.1 If the two CRM values are not within 0.1% RSD, click “Continue” to obtain another repeat until 2 CRM values are within 0.1% RSD.
 - 11.26.2 If the desired precision is not reached, click “Cancel” to escape to troubleshoot the instrument.
- 11.27 Run an additional standard as a sample to ensure the calibration is correct before proceeding to the analysis of samples.
 - 11.27.1 The CRM must be accurate to within 4 μ M, otherwise the system is not precise or accurate and samples should not be analyzed. Ideally, the accuracy of the CRM should be as close to 0 μ M to ensure that any subsequent CRM analyses do not exceed the ± 4 μ M limit for QA/QC.**
 - 11.27.2 If the CRM is accurate to within ± 4 μ M, the calibration is complete.
 - 11.27.3 If the CRM is not accurate to within ± 4 μ M, repeat CRM to determine if the system is precise.
 - 11.27.4 If the repeated CRM are precise but not accurate, recalibrate the HCl acid.
 - 11.27.5 If the repeated CRM are not precise, troubleshoot the instrument using sodium carbonate solution (2300 μ M) to determine the cause of precision.

12. Documentation

- 12.1 Field data will be recorded in the Sampling Log (Appendix F).
- 12.2 Calibration data will be recorded in the TA station lab notebook
- 12.3 Field form shall be kept in the designated file cabinet at CCRI office.

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- 12.4 Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 12.5 All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.
- 12.6 All data will be electronically archived and backed up online.

13. Cleaning and Maintenance

- 13.1 Bottle cleanup process:
 - 13.1.1 Submerge bottles and stoppers in a solution of Liquinox and water for several hours.
 - 13.1.1.1 Dilute Liquinox 1:100 on water.
 - 13.1.2 Remove bottles from the Liquinox solution and rinse with tap water three times.
 - 13.1.3 Rinse the interior of the bottle with distilled water three times.
 - 13.1.4 Place the bottles and stoppers on a drying rack to completely air dry.
 - 13.1.4.1 Place the bottles in an inverted position to remove water from the interior.
 - 13.1.5 Place the caps in a position to air dry.
- 13.2 Clean the surface of the titrator with wet and dry tissues before storage or after a long period of use
- 13.3 In case changing the syringes of the titrator is necessary, refer to the instrument user manual Section C7 on page 25.
- 13.4 The pH electrode should be kept in an appropriate solution (KCl).
 - 13.4.1 Check the level of KCl in the electrode and refill as needed.
 - 13.4.2 A pH=4 buffer solution may also be used for conditioning and storage purposes.

14. Troubleshooting

- 14.1 Refer to Section F of the TALK user manual (p. 30) for common troubleshooting.
- 14.2 Excess of air bubbles:
 - 14.2.1 Check that the system temperature is stable.
 - 14.2.2 Check if the syringe is loose.
 - 14.2.2.1 To tighten, hold the syringe, not the water jacket.
 - 14.2.2.2 Do not overtighten.
- 14.3 If in doubt, contact Dr. Courtney (travis.courtney@upr.edu) for assistance.

15. QA/QC

- 15.1 As per QAPP Table 4, seawater total alkalinity should be measured in micromoles per kilogram ($\mu\text{mol/kg}$) with an accuracy of $4 \mu\text{mol/kg}$ ($\pm 20\%$) and a precision of $<20\%$.
 - 15.1.1 The instrument measures samples volumetrically in units of μM so all samples must be converted to $\mu\text{mol/kg}$ for the final report:

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15.1.1.1 $TA (\mu\text{mol/kg}) = TA (\mu\text{M}) / \text{density}_{\text{sample}}(\text{kg/L})$

15.1.1.2 Sample density can be calculated following the equation by Millero & Poisson (1981) as expressed in the Apollo Total Alkalinity Titrator Instruction Manual.

15.2 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

16. References

- 16.1 Dickson AG, Sabine CL, Christian JR. Guide to best practices for ocean CO₂ measurements. North Pacific Marine Science Organization; 2007.
- 16.2 Apollo SciTech LLC. Model AS-ALK3 Total Alkalinity Titrator Instruction Manual v2022.12